

From Newton's Binomial to Coq: A Complete Educational Breakdown of the Conditional Model of Fermat's Last Theorem

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Abstract

This document logically combines three educational LaTeX drafts, the Russian and English versions of an article on the reconstruction of Fermat's reasoning, and the machine-verifiable file `GlobalNormalization.v`. The text is written so that a strong 11th-grade mathematical lyceum graduate can read it carefully. The main goal of the document is not to announce a new unconditional proof of Fermat's Last Theorem, but to carefully and thoroughly explain a conditional model: if a scale constraint $o(n) = 2$ is correctly isolated from the binary structure of Fermat's equation, then a computer-verified logical chain leads to the impossibility of solutions for $n > 2$. The document sequentially explains Newton's binomial, parameterization via m and p , the roots of unity filter, the equation $k^n = kn$, the local p-adic `PadicBracket`, formal verification in Coq, and the intended message for discussion on `cyberforum.ru`.

Keywords: Fermat's Last Theorem; conditional mathematical model; Newton's binomial; symmetric polynomials; roots of unity filter; binary scaling; exponential-linear equation; equation $k^n = kn$; p-adic valuations; p-adic bracket; `PadicBracket`; formal verification; Coq; interactive theorem proving; computer proof checking; mathematical logic; number theory; educational exposition for a mathematical lyceum.

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Preliminary Honest Remark on the Status of the Model

Fermat's Last Theorem (FLT) has already been proven by modern mathematics in the work of Andrew Wiles and Richard Taylor via modular forms and elliptic curves. This text does not examine the standard proof by Wiles, but rather a separate conditional reconstructive model. Therefore, the key distinction must be established immediately:

An unconditional proof asserts FLT without additional hypotheses. **The conditional model** proves an implication: if a explicitly specified hypothesis on binary scaling is satisfied, then FLT follows from it.

In this form, the model is most suitable for discussion on a mathematical or programming forum. It consists of two distinct parts:

1. **The hypothetico-algebraic part:** a scaling parameter $o(n)$ is extracted from the structure of the two terms, which is then linked to the binary value 2.
2. **The formally verifiable part:** after accepting the condition $o(n) = 2$ or the equivalent relation $2^n = 2n$, Coq verifies that a contradiction arises for $n > 2$.

This separation is crucial: computer verification does not replace the mathematical justification of the initial hypothesis, but it guarantees that all subsequent logical transitions within the chosen model are executed rigorously.

What Exactly Has Been Fixed and Expanded in This Version

Compared to the original educational draft, this version makes five clarifications. First, it explicitly separates *real parameterization* from the *integer problem*. This is important because the parameters m, p, l in the algebraic reconstruction can be real numbers, whereas Fermat's Last Theorem speaks of natural numbers x, y, z . Second, the step

$$y^n = 2n l^n$$

is explained as a real factorization of the core, not as a statement about integer divisibility by n . Third, the local function $q(j)$ is separated from the global scale $o(n)$. Fourth, a modular caveat has been added: solutions to congruences modulo q are not solutions to the original equation in integers. Fifth, it explains more precisely what exactly the `GlobalNormalization.v` file formalizes: it verifies the conditional scheme but does not turn the hypothesis $o(n) = 2$ into an unconditional theorem.

For a mathematical lyceum graduate, this is useful as an example of mathematical culture: a strong idea must be written in such a way that its definitions, domain of applicability, conditional spots, and machine-verifiable consequences are clearly visible.

1 Historical Problem Statement

Fermat's Last Theorem states that the equation

$$x^n + y^n = z^n \tag{1}$$

has no solutions in natural numbers x, y, z for integers $n > 2$. For $n = 1$, such solutions are trivial: for example, $1 + 2 = 3$. For $n = 2$, Pythagorean triples exist, for example:

$$3^2 + 4^2 = 5^2.$$

But starting with cubes, the situation changes radically: a cube cannot be decomposed into two cubes, a fourth power into two fourth powers, and so on.

Fermat formulated this statement in the margins of his copy of Diophantus's *Arithmetica*. In the classic Latin formulation, he wrote that he had discovered a "truly marvelous proof," but the margin was too narrow to contain it. The modern proof, published by Wiles in 1995 with a correction via the Taylor-Wiles paper, relies on deep methods involving modular forms, elliptic curves, and Galois representations. These methods could not have been available to Fermat.

Therefore, a historical-mathematical question naturally arises: could Fermat have possessed a short, elementary idea, even if not in a modern form? This document does not claim to have found that exact proof by Fermat. It offers an educational reconstruction of a conditional model, demonstrating how a rigorous logical scheme can emerge from Newton's binomial, the symmetry of the two terms, and a simple comparison of function growth.

2 How a Mathematical Lyceum Graduate Should Read This Document

To understand the text, it is sufficient to have a solid grasp of the following topics:

- powers, roots, and natural numbers;
- Newton's binomial theorem;
- elementary properties of complex numbers;
- derivatives and monotonicity of functions;
- basic concepts of divisibility and prime numbers.

New concepts, such as p-adic valuations, the roots of unity filter, and Coq, will be introduced gradually. An important pedagogical principle: we will distinguish between "algebraic motivation" and "strict formal implication." This prevents the typical mistake of many amateur Fermat proofs, where a beautiful observation is imperceptibly accepted as a fully proven fact.

3 Mapping Between the Article, Educational Text, and Coq File

To avoid getting lost in the notation, it is convenient to keep the following map in mind.

Level	What is written in the article and text	How it is reflected in Coq v13
Real parameterization	$z = m^n + p^n$, $x = m^n - p^n$, where m, p are not required to be integers	sum_diff_from_parameters_R, structure PositiveRealManuscriptData
Integer specialization	If we require $m, p \in \mathbb{Z}$, parity and perfect power conditions emerge	sum_diff_from_parameters_Z, parity_condition_Z, no_parameters_if_parity_violation
Binomial core	The odd part of the binomial is written as $S = nl^n$ over $\mathbb{R}$	real_core_factorization and real_core_factorization_substitutes_y_power
Scale transition	From $y = ol$, $l > 0$, $y^n = 2nl^n$ follows $o^n = 2n$	positive_l_scaling_forces_cover
Local and global scales	$q(j)$ depends on $j = m/p$, and $o(n) = 2$ is set as a global normalization	LocalScaleGlobalNormalization
Modular caveat	A modular congruence does not imply equality in integers	SectionModularRemark, modular_congruence_not_integer_equality
Final conditional part	With $o(n) = 2$ we get $2^n = 2n$, hence $n \in \{1, 2\}$	covers_with_two_characterisation, fermat_last_theorem_via_maximum_coverage

This mapping helps in reading the document correctly. The educational text explains the idea in human language; the article solidifies the mathematical version of the model; and the Coq file verifies only those formal implications that are explicitly written in the code.

4 Parameterization via m and p

Suppose hypothetically there is a solution to Fermat's equation for some $n > 2$:

$$x^n + y^n = z^n.$$

Let us rewrite it as

$$z^n - x^n = y^n. \tag{2}$$

Now, let us introduce the parameters m and p using the system

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \tag{3}$$

Add and subtract these equations:

$$2m^n = z + x, \quad 2p^n = z - x.$$

Formally, this gives

$$m = \sqrt[n]{\frac{z+x}{2}}, \quad p = \sqrt[n]{\frac{z-x}{2}}. \quad (4)$$

Remark 4.1. If m and p are permitted to be real or complex numbers, such a reconstruction is almost always possible after choosing a branch of the root. However, if we require $m, p \in \mathbb{Z}$, additional conditions arise: $z+x$ and $z-x$ must be even, and the halves $(z+x)/2$ and $(z-x)/2$ must be perfect n -th powers. Therefore, it is important to distinguish between real parameterization and integer parameterization in the model.

Substituting (3) into (2), we obtain

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n. \quad (5)$$

This is exactly where the main algebraic part begins.

4.1 Three Domains of Numbers: $\mathbb{R}$, $\mathbb{Z}$, and $\mathbb{F}_q$

One of the main reasons for confusion in FLT discussions is that the same symbolic expression can exist in different numerical domains. In this model, we need to distinguish between three situations.

1. **The Real Level.** Here $m, p, l \in \mathbb{R}$. We can extract roots, introduce scaling parameters, and work with continuous functions. This is the level where it is convenient to write the reconstruction $m = \sqrt[n]{(z+x)/2}$, $p = \sqrt[n]{(z-x)/2}$.
2. **The Integer Level.** Here we additionally require $m, p \in \mathbb{Z}$ or at least $x, y, z \in \mathbb{N}$. Arithmetic conditions arise: parity of $z+x$, parity of $z-x$, the exactness of n -th powers, and sign consistency.
3. **The Modular Level.** Here we consider remainders modulo q . The congruence

$$x^n + y^n \equiv z^n \pmod{q}$$

can be true even when the equality

$$x^n + y^n = z^n$$

in integers is false. Therefore, modular examples by themselves do not serve as counterexamples to FLT and do not refute the real parameterization.

In Coq v13, this separation is made explicitly: real identities are written using the type $\mathbb{R}$, integer conditions using the type $\mathbb{Z}$, and the modular caveat via the `modular_congruent` relation.

Example 4.2 (Why Real Reconstruction is Not Equal to Integer Reconstruction). Let $n = 3$, $z = 2$, $x = 1$. Then formally

$$m = \sqrt[3]{\frac{3}{2}}, \quad p = \sqrt[3]{\frac{1}{2}}$$

exist as real numbers. But there are no integer m, p such that

$$2 = m^3 + p^3, \quad 1 = m^3 - p^3,$$

because $(z+x)/2 = 3/2$ is not a perfect integer cube. This exact type of distinction is reflected in the Coq lemma `no_parameters_for_example`.

5 Newton's Binomial and the Disappearance of Half the Terms

Recall Newton's binomial formula:

$$(A + B)^n = \sum_{r=0}^n \binom{n}{r} A^{n-r} B^r. \quad (6)$$

For the difference, we have

$$(A - B)^n = \sum_{r=0}^n (-1)^r \binom{n}{r} A^{n-r} B^r. \quad (7)$$

Subtracting (7) from (6), we get

$$(A + B)^n - (A - B)^n = \sum_{r=0}^n (1 - (-1)^r) \binom{n}{r} A^{n-r} B^r \quad (8)$$

$$= 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ is odd}}} \binom{n}{r} A^{n-r} B^r. \quad (9)$$

In our case, $A = m^n$, $B = p^n$, thus

$$y^n = 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ is odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r. \quad (10)$$

This formula contains two crucial facts:

1. the even components cancel out;
2. the odd components are doubled.

If we rewrite the odd index as $r = 2i + 1$, then

$$y^n = 2 \sum_{i=0}^k \binom{n}{2i+1} (m^n)^{n-(2i+1)} (p^n)^{2i+1}, \quad (11)$$

where $k = (n - 1)/2$ for odd n and $k = (n - 2)/2$ for even n .

5.1 Where the Factor n Comes From

For each odd index $r = 2i + 1$, the binomial coefficient can be written as

$$\binom{n}{2i+1} = \frac{n}{2i+1} \binom{n-1}{2i}. \quad (12)$$

This means that in real-rational algebra, we can factor out the common multiplier n from the sum:

$$y^n = 2n L_n(m, p), \quad (13)$$

where $L_n(m, p)$ denotes the remaining expression.

Remark 5.1. One must be careful: formula (12) does not mean that every coefficient $\binom{n}{2i+1}$ is always divisible by n in integers. For example, $\binom{6}{3} = 20$, and 20 is not divisible by 6. Here we are talking about factoring out n as a rational multiplier in an algebraic expression. This is permissible for an educational model, but in strict integer theory, such a step would require separate verification.

If we denote $L_n(m, p) = l^n$, we obtain the key form

$$y^n = 2n l^n, \quad (14)$$

and after extracting the n -th root,

$$y = (2n)^{1/n} l. \quad (15)$$

Let us introduce the scaling function

$$o(n) := \frac{y}{l} = (2n)^{1/n}. \quad (16)$$

Formula (16) is not yet a restriction, but rather a definition of the scale within the algebraic model.

5.2 What the Notation $L_n(m, p) = l^n$ Means

The notation $L_n(m, p) = l^n$ should not be read as an assertion that an integer l has already been found. The more cautious meaning is this: after factoring out n , we call the remaining real quantity l^n . That is, l is introduced as a scaling parameter satisfying

$$l^n = L_n(m, p).$$

If $L_n(m, p) > 0$, we can take the positive real branch of the root:

$$l = \sqrt[n]{L_n(m, p)}.$$

If n is odd, a real root also exists for a negative value of $L_n(m, p)$. If n is even, $L_n(m, p) \geq 0$ is required for a real l . Coq v13 uses the safest case $l > 0$, because then one can cancel out l^n without any issues.

Remark 5.2 (Why This Is Important). If the text simply says “the sum is divisible by n ”, the reader might understand this as the integer assertion $n \mid S$. But in this educational model, a weaker and more correct notation over $\mathbb{R}$ is sufficient:

$$S = n l^n.$$

The Coq file formalizes this step exactly like this: `real_core_factorization` means the real factorization `core = n l^n`, not divisibility in the ring of integers.

Example 5.3 (Checking for $n = 6$). For $n = 6$, $m = 1$, $p = 1$, the sum of the odd binomial coefficients is

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

The number 32 is not divisible by 6 in $\mathbb{Z}$. However, over $\mathbb{R}$, we can always write

$$32 = 6 \cdot \frac{16}{3} = 6 l^6,$$

if we set $l^6 = 16/3$. This illustrates why the phrase about real factorization is more accurate than the phrase about integer divisibility.

6 Roots of Unity Filter

Why does the coefficient 2 appear specifically in (10)? At an elementary level, this is evident from the + and – signs. But on a deeper level, it is a special case of the roots of unity filter.

Definition 6.1. The k -th roots of unity are the complex numbers ω satisfying $\omega^k = 1$. All such roots have the form

$$\omega_j = e^{2\pi i j/k} = \cos \frac{2\pi j}{k} + i \sin \frac{2\pi j}{k}, \quad j = 0, 1, \dots, k-1.$$

The main filter formula is as follows:

$$\sum_{j=0}^{k-1} (\omega_j)^s = \begin{cases} k, & k \mid s, \\ 0, & k \nmid s. \end{cases} \quad (17)$$

Proof. If $k \mid s$, then $\omega_j^s = 1$ for all j , and the sum equals k . If $k \nmid s$, let $q = \omega_1^s \neq 1$. Then

$$1 + q + q^2 + \dots + q^{k-1} = \frac{1 - q^k}{1 - q} = \frac{1 - (\omega_1^k)^s}{1 - q} = 0.$$

□

6.1 The Binary Case $k = 2$

For $k = 2$, the roots of unity are 1 and -1 . Then

$$1^s + (-1)^s = \begin{cases} 2, & s \text{ is even,} \\ 0, & s \text{ is odd.} \end{cases} \quad (18)$$

In the binomial difference, the same mechanism is at work: half the terms disappear, and the other half receive a coefficient of 2. Therefore, the binary structure of Fermat's equation $x^n + y^n = z^n$ is naturally linked to the number 2.

6.2 Generalization to k Terms

If we consider a symmetric model with k terms, the filter (17) isolates the coefficient k . Therefore, in the generalized scale picture, we obtain not $2^n = 2n$, but

$$k^n = kn. \quad (19)$$

This formula is not a theorem about all Diophantine equations with k terms. It is the internal scaling equation of this given model. But it is exactly what explains why the binary case $k = 2$ turns out to be special.

7 Local Function $q(j)$ and Global Normalization $o(n) = 2$

In version v13, it is especially important not to confuse two objects: the local function $q(j)$ and the global scale $o(n)$. They are different levels of the model.

Let

$$m = jp, \quad j > 1.$$

Then $m^n = j^n p^n$, and expression (5) takes the form

$$y^n = (p^n)^n ((j^n + 1)^n - (j^n - 1)^n).$$

By Newton's binomial theorem,

$$(j^n + 1)^n - (j^n - 1)^n = 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ is odd}}} \binom{n}{r} (j^n)^{n-r}.$$

Thus, we can introduce the local function $q(j)$ by the condition

$$q(j)^n = 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ is odd}}} \binom{n}{r} (j^n)^{n-r}. \quad (20)$$

The point $j = 1$ corresponds to $m = p$, which means

$$x = m^n - p^n = 0.$$

For a nontrivial positive solution, this is a degenerate boundary, not a working point. Therefore, inside the working area $j > 1$, the function $q(j)$ generally varies with j . In particular, it is not claimed that $q(j) = 2$ for all $j > 1$.

However, we can consider the boundary value as $j \rightarrow 1+$. Then from (20) we obtain

$$\lim_{j \rightarrow 1+} q(j)^n = 2 \sum_{\substack{1 \leq r \leq n \\ r \text{ is odd}}} \binom{n}{r}.$$

Using the well-known identity for the sum of odd binomial coefficients,

$$\sum_{\substack{1 \leq r \leq n \\ r \text{ is odd}}} \binom{n}{r} = 2^{n-1},$$

we have

$$\lim_{j \rightarrow 1+} q(j)^n = 2^n, \quad \lim_{j \rightarrow 1+} q(j) = 2$$

for the positive branch of the root.

The main clarification in v13: $q(j) \not\equiv o(n)$. The local function $q(j)$ depends on j , while the global normalization chooses the boundary value 2 as the uniform scale of the two-term structure.

That is exactly why the article uses the formulation:

$$o(n) := \lim_{j \rightarrow 1+} q(j) = 2.$$

This is not a pointwise equality $q(j) = 2$, but a normalization step. In Coq v13, this is represented by the `LocalScaleGlobalNormalization` structure: it stores the condition $j > 1$, the boundary equation `boundary_q_equation n 2`, and a separate field `lsgn_global_normalization : lsgn_o = 2`.

8 The Binary Scaling Condition

From formula (16) it follows that

$$o(n)^n = 2n.$$

The core condition of the model asserts that the binary structure of decomposition into two powers fixes the scale as

$$o(n) = 2. \quad (21)$$

Then

$$2^n = 2n. \quad (22)$$

Theorem 8.1 (Conditional Theorem of the Model). *If every hypothetical natural counterexample to FLT for $n > 2$ entails the scaling relation $o(n)^n = 2n$, and if the binary structure of the equation correctly fixes $o(n) = 2$, then the equation $x^n + y^n = z^n$ has no natural solutions for $n > 2$.*

Proof. Given $o(n) = 2$, we obtain $2^n = 2n$. Below it will be proven that the positive integer solutions to this equation are only $n = 1$ and $n = 2$. Therefore, a contradiction arises for $n > 2$. $\square$

8.1 Strict Formulation of the Conditionality

It is convenient to write down the central status of the model as a single implication:

$$\begin{aligned} \text{hypothetical solution for } n > 2 &\implies o(n)^n = 2n, \\ o(n) = 2 &\implies 2^n = 2n \implies n \in \{1, 2\}. \end{aligned}$$

Consequently, for $n > 2$ there are no solutions within the framework of the global binary normalization hypothesis.

That is, the debatable and substantively deep point is not the final relation $2^n = 2n$. The final relation is elementary and machine-verified. The main question is how convincingly the transition from the local two-term structure to the global normalization $o(n) = 2$ is justified. In this educational version, it is treated as an explicitly stated hypothesis of global binary normalization.

9 Exponential vs. Linear Function

Consider the equation

$$2^n = 2n.$$

Divide both sides by 2:

$$2^{n-1} = n. \quad (23)$$

Define the function

$$g(x) = 2^{x-1} - x. \quad (24)$$

Let's check the first values:

$$g(1) = 2^0 - 1 = 0,$$

$$g(2) = 2^1 - 2 = 0.$$

Now look at the derivative:

$$g'(x) = 2^{x-1} \ln 2 - 1. \quad (25)$$

For $x \geq 2$, we have $2^{x-1} \geq 2$, hence

$$g'(x) \geq 2 \ln 2 - 1 \approx 1.386 - 1 = 0.386 > 0.$$

Therefore, on $[2, \infty)$, the function $g(x)$ strictly increases. Since $g(2) = 0$, for all $x > 2$ we have $g(x) > 0$, meaning

$$2^{n-1} > n \quad \text{and} \quad 2^n > 2n \quad \text{for} \quad n > 2. \quad (26)$$

Corollary 9.1. *The equation $2^n = 2n$ in positive integers has exactly two solutions: $n = 1$ and $n = 2$.*

10 Generalization of $k^n = kn$ and the Uniqueness of the Square

Consider

$$k^n = kn \iff k^{n-1} = n. \quad (27)$$

Let

$$g_k(x) = k^{x-1} - x.$$

Then

$$g'_k(x) = k^{x-1} \ln k - 1.$$

If $k \geq 3$ and $x \geq 1$, then

$$g'_k(x) \geq \ln 3 - 1 > 0.$$

Thus, g_k strictly increases right from $x = 1$. Since

$$g_k(1) = 1 - 1 = 0,$$

there are no other positive roots.

Proposition 10.1. *For any integer $k \geq 3$, the equation $k^n = kn$ has only the solution $n = 1$ in positive integers.*

Remark 10.2. This does not mean that equations with three or more terms in actual number theory lack solutions at $n = 2$ or even at some higher powers. For example, sums of three squares equaling a square do exist. The formula $k^n = kn$ describes the internal scale balance of this model, not the entirety of Diophantine reality. This is precisely why the document calls the model conditional.

11 Modular Caveat: Why Modular Congruences Do Not Refute the Model

In discussions about FLT, examples like this are often brought up:

$$a^n + b^n \equiv c^n \pmod{q}.$$

Such examples might be perfectly correct, but they belong to a different level of the problem. A modular congruence means that the difference is a multiple of q :

$$c^n - (a^n + b^n) = qt$$

for some integer t . Equality in integers is only the special case where $t = 0$.

Example 11.1. We have

$$1^3 + 1^3 = 2, \quad 3^3 = 27.$$

In integers, $2 \neq 27$. But modulo 5,

$$27 = 2 + 5 \cdot 5,$$

which means

$$3^3 \equiv 1^3 + 1^3 \pmod{5}.$$

This is an example of a true modular congruence that is not a true integer equality.

The Coq lemma `modular_congruence_not_integer_equality` formalizes this exact logic. Therefore, the existence of solutions to congruences

$$x^n + y^n \equiv z^n \pmod{q}$$

for $n > 2$ is not a counterexample either to Fermat's Last Theorem or to the real parameterization. It is simply a different mathematical object.

If we purposefully switch to modulo q , the parameters must be set separately:

$$m^n \equiv A \pmod{q}, \quad p^n \equiv B \pmod{q},$$

and only then we can write

$$z \equiv A + B \pmod{q}, \quad x \equiv A - B \pmod{q}.$$

In Coq v13, this is reflected by the `ModularParameterResidues` record.

12 p-adic Valuations: Divisibility as a Microscope

A p-adic valuation measures how many times a prime number p appears in the factorization of a number.

Definition 12.1. For a nonzero integer a , the p-adic valuation $v_p(a)$ is the maximum integer $r \geq 0$ such that $p^r \mid a$. For a rational number a/b , it is defined as

$$v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b).$$

Example 12.2. Since

$$24 = 2^3 \cdot 3,$$

we have

$$v_2(24) = 3, \quad v_3(24) = 1, \quad v_5(24) = 0.$$

Basic rules:

$$v_p(ab) = v_p(a) + v_p(b), \tag{28}$$

$$v_p(a + b) \geq \min\{v_p(a), v_p(b)\}. \tag{29}$$

In an educational interpretation, p-adic valuations are like microscopes: a 2-adic microscope sees how many twos are hidden in a number, a 3-adic one sees how many threes, a 5-adic one sees how many fives, and so on.

13 The p-adic Bracket PadicBracket

13.1 Why the Word “Bracket” is Used Here

The term “**bracket**” is used here in its structural sense: as a clamp, a vise, or a constraint. In advanced mathematics and theoretical physics, this word often denotes special mathematical operations that place variables into strict boundaries. For example, in analytical mechanics, **Poisson brackets** are used, and in quantum theory and the theory of constraints, **Dirac brackets**. Such brackets are not merely “drawn around an expression,” but they define a rule that permits certain motions or states while forbidding others.

In this document, the name PadicBracket should be understood in this figurative sense: it is a **p-adic lock**, or a **p-adic vise**. This section of the Coq code takes a continuous real scaling parameter o , which in analysis could be fractional or irrational, and subjects it to testing from the perspective of discrete arithmetic: which prime numbers can enter its local structure, and which must disappear. In other words, the continuous parameter is “clamped” within the constraints of divisibility theory.

13.2 The Idea of p-adic “Probing” in Layman’s Terms

When we write the real equality

$$o^n = 2n,$$

we are operating in the realm of continuous numbers. In this world, o can be not only an integer but also a rational, irrational, positive real number. For instance, the formula $o^n = 2n$ formally implies

$$o = (2n)^{1/n}.$$

For an arbitrary n , this is generally not an integer. Therefore, real analysis alone is insufficient if we want to understand the model’s connection to natural solutions of Fermat’s equation.

But Fermat’s equation seeks solutions specifically in natural numbers. In the world of natural and integer numbers, the **Fundamental Theorem of Arithmetic** applies: every natural number can be uniquely factored into prime numbers. Thus, a natural move arises: to view the scaling parameter not only as a real quantity but also to “probe” it locally via prime numbers.

The idea is very simple. We ask:

- does the parameter o see the prime number 3?
- does it see the prime number 5?
- does it see the prime number 7?
- or does only the binary component remain after local testing?

Mathematically, such “probing” is written via the p-adic valuation $v_p(o)$. It shows to what power the prime number p enters the object we are analyzing. If $v_3(o) = 0$, there is no three in the parameter. If $v_5(o) = 0$, there is no five. If $v_2(o) = 1$, the two enters exactly once.

13.3 How the p-adic Lock is Structured in Coq

A minimal abstraction is introduced in the Coq file:

```
Record OPadic := { vp_o : nat -> Z }.
Definition NatOddGreater1 (p : nat) : Prop := (1 < p)%nat.
Definition padic_equation (o : OPadic) (n : nat) : Prop :=
  forall p, NatOddGreater1 p -> Nat.odd p = true ->
    Z.of_nat n * vp_o o p = 0%Z.
```

Here `OPadic` stores the function $p \mapsto v_p(o)$, i.e., it associates each p with the p-adic valuation of the scaling parameter. The equation `padic_equation` can be read as:

$$n \cdot v_p(o) = 0.$$

It states: for every odd local parameter $p > 1$, the product of the exponent n and the p-adic valuation of o is equal to zero. In familiar high-school logic, this resembles the statement: if the positive factor n is not zero, the only way to get a zero product is to nullify the second part, namely $v_p(o)$.

Remark 13.1. In the provided Coq fragment, the condition `NatOddGreater1 p` means “ $p > 1$ ”, and together with `Nat.odd p = true` it means “ p is odd”. The code itself in this minimal version does not strictly require proving that p is prime separately. Therefore, formally, the lemma nullifies all odd natural local indices $p > 1$, but in the standard p-adic interpretation, the main meaning relates primarily to odd prime numbers $3, 5, 7, \dots$, because prime numbers are the basic “microscopes” of arithmetic.

Next, the following lemma is formulated:

```
Lemma odd_primes_vanish_in_o
  (o : OPadic) :
  padic_equation o 1%nat ->
  padic_equation o 2%nat ->
  forall p, NatOddGreater1 p -> Nat.odd p = true ->
    vp_o o p = 0%Z.
```

The meaning of this lemma is as follows: if the p-adic equation is imposed on the parameter o in the base cases, then for all odd $p > 1$ the corresponding valuation becomes zero. The proof relies on the case $n = 1$:

$$1 \cdot v_p(o) = 0 \implies v_p(o) = 0.$$

This is very simple logic, but having it in a machine-verified form is crucial. In Coq, this exact step appears especially transparent: the compiler takes the equation for the case $n = 1$, rewrites the factor `Z.of_nat 1` as one, and applies the multiplication by one rule `Z.mul_1_l`. After this, the equality

$$1 \cdot v_p(o) = 0$$

automatically transforms into

$$v_p(o) = 0.$$

Substantively, this means: if the p-adic lock already triggers in the linear base case, then no odd local factor can be preserved in the structure of the parameter o . Odd prime numbers $3, 5, 7, \dots$ are “squeezed out” of the parameter.

13.4 Final 2-adic Normalization

The 2-adic normalization is specified separately:

```
Lemma two_adic_normalisation (o : OPadic) :
  Z.of_nat 1 * vp_o o 2%nat = 1%Z ->
  vp_o o 2%nat = 1%Z.
```

This lemma says: if the 2-adic component is normalized by the condition

$$1 \cdot v_2(o) = 1,$$

then after removing the redundant factor of one, we are left with

$$v_2(o) = 1.$$

In everyday language of divisibility, this means that two enters the scaling parameter exactly once. In other words, the p-adic lock first nullifies all odd local components, and then separately fixes the only surviving binary component.

13.5 Summary of the PadicBracket Operation

The operation of the PadicBracket section can be represented as a sequence of questions:

1. Is there a three in the parameter o ? The answer of the local analysis: no, $v_3(o) = 0$.
2. Is there a five? No, $v_5(o) = 0$.
3. Is there a seven and other odd primes? No, the corresponding valuations also drop to zero.
4. What is left? Only the binary component is left, which is fixed by the condition $v_2(o) = 1$.

Therefore, PadicBracket is a useful convergence point between two worlds:

- **the continuous world of analysis**, where the real formula $o^n = 2n$ emerges;
- **the discrete world of arithmetic**, where every numerical object is tested via its prime divisors.

In a strong intuitive formulation, one could say: the p-adic bracket shows that if the scaling parameter has to pass a local arithmetic check, no odd prime components survive, while the binary structure remains central. This is why this block supports the idea of binary scaling $o(n) = 2$ not just as a simple heuristic, but as an arithmetically motivated normalization within the conditional model.

Remark 13.2. It is important to understand the status of this block. PadicBracket is not an independent, complete proof that the real parameter o unconditionally equals 2. It formalizes a local scheme: under given p-adic equations, odd prime components vanish, and the 2-adic component is fixed separately. Therefore, this section strengthens the algebraic interpretation of binarity, but it remains part of the conditional model.

14 Connection with the Main Fragments of Coq v13

The file `GlobalNormalization.v` is structured not as a long text, but as a set of small, verifiable statements. It is useful for a student to see the mathematical meaning behind their names.

Name in Coq	Mathematical meaning
<code>sum_diff_from_parameters_R</code>	If $z = m^n + p^n$, $x = m^n - p^n$, then $z + x = 2m^n$, $z - x = 2p^n$ over $\mathbb{R}$.
<code>parameter_equations_raise_nat</code>	If the equations for z and x are defined at the natural level, they can be raised to the power n .
<code>real_core_factorization</code>	The binomial core is written as $\text{core} = nl^n$ over $\mathbb{R}$, without claiming integer divisibility.
<code>positive_l_scaling_forces_cover</code>	From $l > 0$, $y = ol$, $y^n = 2nl^n$ follows $o^n = 2n$.
<code>covers_with</code>	The predicate $o^n = 2n$.
<code>covers_with_two_characterisation</code>	If $2^n = 2n$, then $n = 1$ or $n = 2$.
<code>fermat_last_theorem_via_maximum_coverage</code>	Conditional form of FLT: if any hypothetical counterexample yields $2^n = 2n$, then there are no counterexamples for $n > 2$.
<code>ModularRemark</code>	Formal separation of modular congruences from integer equalities.
<code>PadicBracket</code>	A minimal local p -adic scheme showing the vanishing of odd local components given p -adic conditions.

This table is pedagogically important. It shows that Coq does not “understand Fermat” or “guess historical intent”. Coq only verifies what is formally written. That is exactly why one must honestly state in the text: global binary normalization $o(n) = 2$ remains a structural hypothesis of the model.

15 What Exactly Coq Verifies

Coq is an interactive theorem proving system. In regular programming, code is written so the machine executes it. In Coq, code is written so the machine verifies a proof. If a proof contains a logical gap, the file will not compile.

The central definition in the file is:

```
Definition covers_with (o : R) (n : nat) :=
  pow o n = 2 * INR n.
```

It says: a parameter o “covers” an exponent n , if

$$o^n = 2n.$$

Next, it is proven that for $o = 2$ this is only possible when $n = 1$ or $n = 2$:

```
Lemma covers_with_two_characterisation (n : nat) :
  covers_with 2 n -> n = 1%nat \/ n = 2%nat.
```

Then the conditional variant of FLT is formulated:

```

Corollary fermat_last_theorem_via_maximum_coverage :
  (forall (n x y z : nat),
    (2 < n)%nat ->
      (Nat.pow x n + Nat.pow y n)%nat = Nat.pow z n ->
        pow 2 n = 2 * INR n) ->
  forall (n x y z : nat),
    (2 < n)%nat ->
      (Nat.pow x n + Nat.pow y n)%nat = Nat.pow z n -> False.

```

Let's read this in English:

If for any hypothetical solution to Fermat's equation with $n > 2$, the scaling equality $2^n = 2n$ holds, then any such solution yields False, i.e., a contradiction.

In other words, Coq does not verify the historical guess about the origin of $o(n) = 2$, but rather the strict implication:

(binary scaling hypothesis) $\implies$ (impossibility of solutions for $n > 2$).

16 What This Document and the Coq File Do Not Prove

For a proper public discussion, it is vital to explicitly state the boundaries of the result.

1. **It is not proven unconditionally that $o(n) = 2$.** This is the hypothesis of global binary normalization. In the article, it is motivated by the algebraic roots of unity filter and the boundary value $\lim_{j \rightarrow 1+} q(j) = 2$, but in Coq it is not derived as an unconditional theorem solely from Fermat's equation.
2. **A complete theory of finite fields is not built.** The modular section is only needed to show that a modular congruence is weaker than an equality in $\mathbb{Z}$.
3. **The complete analytical limit $\lim_{j \rightarrow 1+} q(j) = 2$ is not formalized.** In Coq, it is reflected through the boundary equation and the normalization structure, rather than an elaborate theory of limits.
4. **Wiles's proof is not replaced.** Wiles's proof is an unconditional theorem of modern mathematics. This text is an educational reconstruction of a conditional model.

These limitations do not diminish the educational value of the document. On the contrary, they make it mathematically more honest: a student sees not only a beautiful idea but also the boundary between a proven statement, a hypothesis, and motivation.

17 Why the 4.0 Seconds Compilation Time is Important for Programmers

In discussions on a programming forum, it is important to emphasize not only the mathematical idea but also its reproducibility. The command

```
coqc GlobalNormalization.v
```

finishes the verification in about 4.0 seconds. This means:

- the file contains no unfinished proofs like `Admitted`;
- tactics like `lia`, `lra`, `ring` cleanly close the corresponding arithmetic and algebraic subgoals;
- the verified part is compact enough and reproducible;
- any forum member can download Coq and verify the file on their own.

For mathematicians, this does not replace peer-reviewing the hypothesis, but for programmers, it is a strong argument: the logical part of the model is not just a bunch of words; it has been reduced to code that can be run.

18 How to Correctly Formulate a Post for cyberforum.ru

Below is a detailed version of the message, built on the source text, but included in the document as a methodological template. It can be used as a basis for a post, keeping the main points: a respectful tone, explicit acknowledgment of the model's conditional nature, and an emphasis on verifiability.

Topic: Radical Update of the Model: Coq, p-adic brackets, and the roots of unity filter

Greetings to all forum members!

Our discussion in this section has been going on for a long time, and my work has seriously evolved since then. I am grateful to the platform for the opportunity to freely discuss unconventional mathematical ideas. At the same time, I perfectly understand why academic math forums are extremely harsh toward threads concerning Fermat's Last Theorem: too often, under the guise of "elementary proofs", people present texts with implicit division by zero, erroneous leaps from real numbers to integers, or unverified generalizations.

That is why in this new version I propose not to discuss the slogan "FLT is proven by high-school methods," but a more precise and testable formulation: a conditional model has been built, in which the binary scaling hypothesis is explicitly separated from those consequences that have already been verified by a machine in Coq.

The key updates are as follows.

1. **Algebraic depth instead of a mere heuristic.** The choice of the scaling parameter $o(n) = 2$ is now explained through the binary structure of the equation $x^n + y^n = z^n$ and the roots of unity filter. When subtracting binomials $(m^n + p^n)^n - (m^n - p^n)^n$, the even components cancel out, and the odd components receive a coefficient of 2. In a generalized symmetric picture for k terms, a similar filter extracts the coefficient k . For FLT, the case $k = 2$ is fundamentally crucial.

2. **Generalization of $k^n = kn$.** The internal scaling equation of the model has the form $k^n = kn$. For $k \geq 3$, it has only the trivial positive integer root $n = 1$. For $k = 2$, exactly two roots appear: $n = 1$ and $n = 2$. This beautifully explains the special position of the quadratic case and Pythagorean triples within this model.
3. **Local p-adic bracket.** A PadicBracket section has been added to the file. The word “bracket” is best understood here as an “arithmetic clamp” or “p-adic lock”: the code block translates the real scaling parameter o into the language of local valuations $v_p(o)$. Under given local equations, odd prime components vanish, while the 2-adic component is fixed separately. This should not be passed off as an independent unconditional proof of FLT, but it is a useful formal layer that connects the binary nature of the model with the arithmetic of prime numbers.
4. **Complete verification in Coq.** The entire logical part, starting from the definition of `covers_with` and ending with the conditional corollary `fermat_last_theorem_via_maximum_coverage`, complies with the command `coqc GlobalNormalization.v`. In the current version, the verification takes about 2.0 seconds. This means the conditional implication is strictly written and can be independently verified.

The point of the new version is this: I am not asking you to take an unconditional proof of FLT on faith. I am proposing to verify a specific formal model. In it, one can clearly see where the hypothesis lies, where the algebraic motivation is, and where the machine-verified consequence begins. I would be grateful for substantive criticism of exactly these spots: the correctness of the transition to $o(n) = 2$, the p-adic interpretation, and the structure of the Coq code.

19 Frequently Asked Questions

19.1 Does this document unconditionally prove FLT?

No. The document proves a conditional implication: if the binary scaling hypothesis is correct, then FLT follows from it. An unconditional proof of FLT already exists by Wiles, but it is of an entirely different nature.

19.2 Then why is Coq needed?

Coq is needed to remove doubts in the part that is pure logic. It does not prove the initial hypothesis for us, but it checks that after accepting it, the derivation to a contradiction at $n > 2$ contains no logical errors.

19.3 Why is the quadratic case special?

Because in the equation $2^n = 2n$, the positive integer solutions are only 1 and 2. The case $n = 2$ corresponds to Pythagorean triples. For $n > 2$, the exponent 2^n grows faster than the linear function $2n$.

19.4 Why can't we just say that binary scale is obvious?

Because this exact step is the main mathematical crux. It must be separately justified through the two-term structure, the roots of unity filter, and the p-adic interpretation. If this step is not proven in a fully unconditional sense, the model remains conditional.

19.5 Why does the text separately discuss modular arithmetic?

Because an equality always implies a modular congruence, but a modular congruence does not imply an equality. If

$$x^n + y^n = z^n,$$

then certainly

$$x^n + y^n \equiv z^n \pmod{q}$$

for any q . But the reverse move is invalid. A modular congruence only indicates that the difference is a multiple of q . Therefore, modular examples do not close or refute the real scheme; they belong to a separate level of analysis.

19.6 Why can't $q(j)$ simply be replaced by 2?

Because $q(j)$ is a local function of $j = m/p$. For $j > 1$, it generally depends on j . The value 2 appears as a boundary normalization as $j \rightarrow 1+$. Thus, the correct v13 notation is:

$$q(j) \not\equiv o(n), \quad o(n) := \lim_{j \rightarrow 1+} q(j) = 2.$$

These are different levels: a local scale and a global normalization.

19.7 Why is real factorization not divisibility?

The real formulation $S = nl^n$ means that we define $l^n = S/n$. Integer divisibility $n \mid S$ implies much more: that S/n is an integer. The model uses the former, weaker and more correct, notation. This is exactly what the Coq predicate `real_core_factorization` formalizes.

20 Educational Exercises

1. Expand $(A + B)^5 - (A - B)^5$ and ensure that only odd powers of B remain.
2. Prove the roots of unity filter formula for $k = 3$ directly, using $1 + \omega + \omega^2 = 0$.
3. Verify manually that $2^n = 2n$ has solutions $n = 1, 2$, but no solutions for $n = 3, 4, 5$.
4. Prove by induction that $2^n > 2n$ for all $n \geq 3$.
5. Find $v_2(40)$, $v_5(40)$, $v_3(40)$.
6. Explain in your own words why Coq verifies a conditional theorem, rather than a historical hypothesis about what exactly Fermat had in mind.
7. For $n = 6$, calculate $\binom{6}{1} + \binom{6}{3} + \binom{6}{5}$ and explain why this is a good example of the difference between real factorization and integer divisibility.
8. Find an example of numbers a, b, c, q for which $a^3 + b^3 \equiv c^3 \pmod{q}$, but $a^3 + b^3 \neq c^3$ in integers.
9. Explain why the condition $j > 1$ makes the point $j = 1$ a boundary point, rather than a working point of the parameterization.

21 Final Scheme of the Entire Model

The whole logic of the document can be compressed into the following chain:

$$\begin{aligned}
 x^n + y^n &= z^n \implies z^n - x^n = y^n, \\
 z = m^n + p^n, \ x = m^n - p^n &\implies y^n = (m^n + p^n)^n - (m^n - p^n)^n, \\
 \text{Newton's binomial} &\implies y^n = 2n l^n, \\
 \text{scaling function} &\implies o(n)^n = 2n, \\
 \text{binary constraint} &\implies o(n) = 2, \\
 &\implies 2^n = 2n, \\
 \text{growth analysis} &\implies n \in \{1, 2\}, \\
 n > 2 &\implies \text{contradiction.}
 \end{aligned}$$

The strictly verified part in Coq starts not with historical intuition, but with the formal condition $2^n = 2n$ for a hypothetical solution. From that point onward, everything truly becomes ironclad logic.

22 Conclusion

The combined version demonstrates how one and the same idea unfolds in several languages:

- in the language of high-school algebra — via Newton's binomial;
- in the language of complex numbers — via roots of unity;
- in the language of calculus — via the comparison of an exponential and a linear function;
- in the language of arithmetic — via p-adic valuations;
- in the language of formal logic — via Coq.

For an 11th-grade student at a mathematical lyceum, this is valuable not only as a story about Fermat's Last Theorem but also as an example of modern proof culture. In the 21st century, a mathematical idea must be able to live in three forms simultaneously: in a human explanation, in a strict formula, and in testable code. The current conditional model is valuable precisely because it makes the boundaries of these three forms explicit.

A Mini-Glossary

Term	Meaning
FLT	Fermat's Last Theorem: $x^n + y^n = z^n$ has no natural solutions for $n > 2$.
Newton's binomial	The formula for expanding $(A + B)^n$ using binomial coefficients.
Roots of unity filter	The sum of powers of k -th roots of unity, which yields k or 0.
Scaling parameter $o(n)$	An internal parameter of the model, satisfying $o(n)^n = 2n$.
Binary constraint	The hypothesis that the two-term structure of FLT fixes $o(n) = 2$.
p-adic valuation	The exponent of the prime p in the prime factorization of a number.
Bracket	In this document, a figurative name for the p-adic block that clamps the scaling parameter with local divisibility conditions.
Coq	An interactive theorem proving system.
Local function $q(j)$	A binomial scale depending on the ratio $j = m/p$. In the working area $j > 1$, it is not necessarily equal to 2.
Global normalization	A separate step of the model in which the boundary value $\lim_{j \rightarrow 1+} q(j) = 2$ is used to select $o(n) = 2$.
Real factorization	An expression of the form $S = nl^n$ over $\mathbb{R}$; not the same as integer divisibility $n \mid S$.

B List of References and Guidelines

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